

WEEKLY JOURNAL DIGEST

Weekly journal digest for 2026-06-08 to 2026-06-14

Curated articles in political communication and computational social science.

SUMMARY

COMAP Journal Bot reviewed 244 collected candidates for the 2026-06-08 to 2026-06-14 digest window. After manual relevance filtering, the curated digest keeps 41 papers: 23 new this week, 18 previous-week catch-up items, and 0 late additions.

This is a strong week for computational media and public-interest measurement. The new articles cluster around algorithmic identity work, misinformation referrals across platforms, YouTube pathway auditing, LLM reporting standards, migration and mobility reconstruction, and legal-discourse shifts around LGBTQ rights. The general-science stream was noisy, but it contributed several useful pieces on AI infrastructure, urbanization, climate governance, and social determinants of health.

- Platform governance is the center of gravity: misinformation referrals, YouTube radicalization pathways, algorithmic media experiences, and media distrust all appear in the curated set.
- AI-for-CSS work is practical rather than speculative this week, with papers on LLM reporting standards, open-source AI for sustainable development, and deep-learning migration measurement.
- Policy-facing social science remains broad: international trust, voting, religious messages, climate targets, food adoption barriers, urbanization, and public-health behavior all make the cut.

HIGHLIGHTS

When the Algorithm Understands You (or Doesn't): LGBTQ+ Youth's Identity-Related Interactions with Algorithmic Media

COMMUNICATION RESEARCH · 2026-06-13

Y. Anthony Chen; Samuel Hardman Taylor

Why it matters · This paper puts algorithmic responsiveness into the lived terrain of identity formation, giving COMAP readers a sharper vocabulary for how recommendation systems can validate, misrecognize, or reshape marginalized users' media experiences.

[Open article →](#)

Transferable human mobility network reconstruction with neuroGravity

NATURE COMPUTATIONAL SCIENCE · 2026-06-12

Jinming Yang; Shaoyu Huang; Zongyuan Huang; Yaohui Jin; Xiaokang Yang; et al. (7 total)

Why it matters · A strong computational-infrastructure paper for urban planning and public health: it reconstructs mobility networks where travel-survey data are missing, which is exactly the kind of measurement gap social scientists routinely face.

[Open article →](#)

Steering open-source AI to accelerate the sustainable development goals

NATURE COMMUNICATIONS · 2026-06-11

Min Chen; Kai Wu; Prajal Pradhan; Cameron Allen; Stefano Nativi; et al. (21 total)

Why it matters · The article connects open-source AI to governance and sustainable-development capacity, making it useful for thinking beyond model performance toward institutional deployment and public-interest infrastructure.

[Open article →](#)

Deep learning four decades of human migration

NATURE · 2026-06-10

Thomas Gaskin; Guy J. Abel

Why it matters · This is a high-value migration measurement paper: it uses deep learning to map four decades of human movement and may be useful for anyone tracking population change with incomplete official data.

[Open article →](#)

Decriminalizing same-sex relationships and its impact on social media discussions in Asia

NATURE HUMAN BEHAVIOUR · 2026-06-10

Kokil Jaidka; Preetika Verma; Yiting Chen

Why it matters · The study is a direct fit for digital public-sphere work, linking legal change around same-sex relationships to social media discussion across Asian contexts.

[Open article →](#)

Evaluating Echo Chambers, Rabbit Holes, and Radicalization Pathways on YouTube

POLITICAL COMMUNICATION · 2026-06-10

Megan A. Brown; James Bisbee; Angela Lai; Richard Bonneau; Jonathan Nagler; et al. (6 total)

Why it matters · A core platform-politics contribution: it evaluates whether YouTube use empirically produces echo chambers, rabbit holes, or radicalization pathways rather than assuming those mechanisms exist.

[Open article →](#)

A reporting checklist for large language models in behavioural science

NATURE HUMAN BEHAVIOUR · 2026-06-09

Stefan Feuerriegel; Christopher Barrie; M. J. Crockett; Laura K. Globig; Killian L. McLoughlin; et al. (81 total)

Why it matters · This checklist gives behavioral scientists a concrete reporting standard for LLM-based research, which is increasingly necessary for reproducible AI-assisted computational social science.

[Open article →](#)

Multi-Platform Referrers of Misinformation: A Comparative Analysis of Misinformation Visits Referred by Facebook, Twitter, Instagram, Reddit, YouTube, Snapchat, and TikTok

POLITICAL COMMUNICATION · 2026-06-09

Ross Dahlke; Ryan C. Moore; Danya Adib-Azpeitia; Johan Ugander; Jeffrey T. Hancock

Why it matters · This is one of the most COMAP-relevant items this week: it compares how major platforms refer visits to misinformation, moving the evidence base beyond single-platform claims.

[Open article →](#)

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Previous Week Catch-Up

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COLLECTION SNAPSHOT		
TOTAL COLLECTED CANDIDATES 244	NEW THIS WEEK CANDIDATES 158	PREVIOUS WEEK CATCH-UP CANDIDATES 86
LATE ADDITIONS CANDIDATES 0	CURATED DIGEST BELOW 23 new, 18 catch-up, 0 late additions	NOTE The full collected set above is before manual relevance filtering, while the HTML email body, attached full HTML artifact, and attached PDF reflect the curated digest.

FULL CURATED DIGEST

New This Week

Communication Research

Co-Opetition in Crowdsourcing Challenges: Comment Network Dynamics and Relational Homophily Among Contestants in Crowdsourcing Challenge Communities

2026-06-13 · DOI 10.1177/00936502261458941

Yiqi Li; Peter Monge; Janet Fulk

Syracuse University, NY, USA; University of Southern California, Los Angeles, CA, USA

Abstract. Situated in a creative crowdsourcing challenge organizational community, this research examines how different types of homophily affect network formation among ideators, the challenge innovators. Longitudinal analysis of comment network events among ideators was conducted with relational event models. Status homophily was examined based on ideators' self-reported public profiles. Findings show that ideators' location by country significantly influences relational events, but occupational homophily did not. Value homophily was examined based on the content similarity of ideators' idea submissions. Results indicate that ideator's choices of platform content categories, those that are designed by the challenge management, significantly influenced ideators' communication patterns. Specifically, long-term needs were a salient homophily factor among platform categories, whereas short-term needs were heterophilous factors. Finally, a multidimensional homophily measurement using latent class analysis on idea-category-submission patterns predicted the occurrences of communication relational events and achieved an improved model fit compared to modeling all platform categories together.

[Read article →](#)**When the Algorithm Understands You (or Doesn't): LGBTQ+ Youth's Identity-Related Interactions with Algorithmic Media**

2026-06-13 · DOI 10.1177/00936502261459750

Y. Anthony Chen; Samuel Hardman Taylor

Stanford University, CA, USA; University of California Irvine, Stanford, USA; University of Illinois Chicago, USA

Abstract. During their daily media use, youth interact with algorithms. Drawing on the algorithm responsiveness process framework, this paper examines the co-construction of identity between youth and algorithmic media, focusing on perceived algorithm responsiveness (PAR), perceived algorithm insensitivity (PAI), and algorithm training as sociotechnical processes that structure how LGBTQ+ youth perceive themselves. Study 1 (N = 335), using six weekly diaries among LGBTQ+ youth, showed that affirming LGBTQ+ representations curated by algorithms predicted higher PAR and lower PAI, especially when less algorithm training was needed. In turn, higher PAR and, unexpectedly, higher PAI predicted greater identity pride. Study 2 (N = 664), a cross-sectional survey, found that LGBTQ+ youth reported lower PAR than non-LGBTQ+ youth. An indirect effect of algorithm training on self-concept clarity via PAR emerged for non-LGBTQ+ participants but not for LGBTQ+ participants. These findings highlight social identity in algorithmic media as a relational, dynamic process, and LGBTQ+ youth can shape content recommendations for identity-related needs.

[Read article →](#)**Two Forms of Media Distrust: Cynicism, Skepticism, and Online News Exposure**

2026-06-10 · DOI 10.1177/00936502261458955

Lukas P. Otto; Frank Mangold; Michaela Maier

GESIS - Leibniz Institute for the Social Sciences, Cologne, Germany; RPTU Kaiserslautern-Landau, Institute for Communication Psychology and Media Education, Germany

Abstract. The relationship between trust in the news media and news use has been studied extensively, yet findings remain inconsistent. We argue that a key source of this inconsistency is the conflation of two distinct forms of media distrust: skepticism and cynicism. Additionally, prior research largely relies on self-reported news use, which is prone to bias. To address both issues, we link survey data with web-tracking data from 739 participants during the 2021 German federal election. We show that skeptics and cynics differ in personality traits, political orientations, and news-use motivations, and - crucially - in their actual news consumption behavior. While skepticism is associated with broader and more intense news exposure, cynicism is linked to lower exposure, particularly to established journalistic mainstream media. These results demonstrate that different forms of media distrust are associated with distinct patterns of news exposure, underscoring the importance of conceptual and behavioral precision in media trust research.

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Nature Communications

A probabilistic forecasting framework for neighbourhood-level disaggregation of electric vehicle adoption scenarios

2026-06-13 · DOI 10.1038/S41467-026-74155-0

Isaac Flower; Furong Li; Julian Padget

Abstract. Abstract The rapid growth of electric vehicle (EV) adoption presents significant challenges for electricity networks, particularly at the low-voltage level, where clustered neighbourhood demand risks overloading infrastructure. Existing scenario-based planning approaches typically assume uniform EV uptake across neighbourhoods within a region, failing to capture the heterogeneity in historical EV registration data. They provide limited uncertainty quantification, despite the difficulty of predicting future adoption at fine spatial scales. This paper introduces a Gaussian process (GP)-based forecasting framework that combines granular historical EV registration data with top-down regional scenarios to generate probabilistic neighbourhood-level forecasts. The GP captures how local adoption deviates from regional trends, encoded in the GP's mean function, ensuring consistency with broader scenarios while accounting for local variation and uncertainty. We validate the framework using ten representative local authority districts in England and Wales, covering 1,294 neighbourhoods. The framework demonstrates improved performance compared to baseline methods (scaled scenario, logistic growth, linear extrapolation) in normalised mean absolute error, with statistically significant improvements at horizons of three years and beyond. It also delivers well-calibrated prediction intervals, providing reliable uncertainty estimates. This framework offers a practical tool for network operators, policymakers, and planners to support targeted decision-making and investment.

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Steering open-source AI to accelerate the sustainable development goals

2026-06-11 · DOI 10.1038/S41467-026-73866-8

Min Chen; Kai Wu; Prajal Pradhan; Cameron Allen; Stefano Nativi; Klaus Hubacek; Alexey Voinov; Felix Creutzig; Tatiana Filatova; Niklas Boers; Michael Meadows; Peilong Ma; et al. (21 total)

Abstract. While the artificial intelligence (AI) revolution is advancing rapidly, the open-source paradigm offers key pathways and potential risks for accelerating progress towards the Sustainable Development Goals and beyond. This comment introduces four governance actions that consider how sustainability, evaluation, safety, and cooperation can be integrated into the transformation of open-source AI, thereby reducing uncertainties and challenges posed by open-source AI for sustainable global prosperity. This comment introduces four governance actions that consider how sustainability, evaluation, safety, and cooperation can be integrated into the transformation of open-source AI, thereby reducing uncertainties and

challenges posed by open-source AI for sustainable global prosperity.

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Navigating adoption barriers for microbial proteins in future food

2026-06-10 · DOI 10.1038/S41467-026-73987-0

Rima Gnaim; Hakimi Kassim; Lisa Neidhardt; Thomas Gassler; Rodrigo Ledesma-Amaro

Abstract. Abstract Microbial biomass fermentation, in which microbes are cultivated to produce nutrient-dense biomass, offers a scalable route to sustainable protein production with low land, water, and greenhouse gas footprints. However, its shift from a speciality to a mainstream food continues to be difficult. Here, we move beyond technological overviews and propose a three-phase adoption framework: novelty barrier, early trust-building, and mainstream normalisation. This framework organises techno-economic, regulatory, and infrastructural barriers into a single trajectory. We trace single-cell proteins' rise, decline and resurgence, map engineering and policy enablers by phase, and outline levers to move microbial proteins into resilient food systems.

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Nature Computational Science

Transferable human mobility network reconstruction with neuroGravity

2026-06-12 · DOI 10.1038/S43588-026-01003-Y

Jinming Yang; Shaoyu Huang; Zongyuan Huang; Yaohui Jin; Xiaokang Yang; Marta C. Gonzalez; Yanyan Xu

Abstract. Accurate modeling of human mobility is critical for tackling urban planning and public health challenges. In undeveloped regions, the absence of comprehensive travel surveys necessitates reconstructing mobility networks from publicly available data. Here we develop neuroGravity, a physics-informed deep learning model that reliably reconstructs mobility flows from limited observations and transfers to unobserved cities. Using only urban facility and population distributions, we find that neuroGravity's regional representations strongly correlate with socioeconomic and livability status, offering scalable proxies for costly surveys. Furthermore, we uncover that spatial income segregation has a key role in model transferability: mobility networks are most reliably reconstructed when target cities share similar segregation levels with the source. We design an index to quantify this segregation and accurately predict transferability. Finally, we generate mobility flow proxies for over 1,200 cities worldwide, highlighting neuroGravity's potential to mitigate critical data shortages in resource-limited, underdeveloped areas.

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Nature Human Behaviour

Social determinants of health and epigenetic clocks: a systematic review and meta-analysis of 140 studies

2026-06-12 · DOI 10.1038/S41562-026-02477-6

Y. E. Willems; A. D. Rezaki; M. Aikins; A. Bahl; Q. Wu; D. W. Belsky; L. Raffington

Abstract. Abstract Social determinants of health-such as socioeconomic status (SES) and race and ethnicity-strongly shape health. Individuals with lower SES or marginalized identities experience earlier onset of disease and shorter lifespans. 'Epigenetic clocks' measure biological ageing and are increasingly used to study healthy ageing, yet it remains unclear which clocks are most sensitive to social inequality. Here we conducted an Open Science Framework-pre-registered systematic review and meta-analysis of 140 studies (N = 65,919; 1,065 effect sizes) testing associations of SES and race and ethnicity with three generations of clocks. PubMed, PsycINFO, Web of Science, medRxiv and bioRxiv were searched from February

2024 to September 2025. Eligible studies included empirical articles in English published since 2013 on non-clinical populations reporting at least one relevant association. Associations between SES and biological ageing varied across different generations of epigenetic clocks ($F(2) = 178.10$, $P = -0.03$, 95% confidence interval (CI) [0.04, 0.01], $P = -0.11$, 95% CI [0.12, 0.09], $P = -0.13$, 95% CI [0.15, 0.11], P

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Decriminalizing same-sex relationships and its impact on social media discussions in Asia

2026-06-10 · DOI 10.1038/S41562-026-02485-6

Kokil Jaidka; Preetika Verma; Yiting Chen

Abstract. Abstract While legal reforms may coincide with shifts in public discourse, broader normative change often remains uneven and context dependent. Since 2018, four Asian countries-India, Taiwan, Thailand and Singapore-have decriminalized same-sex relationships, providing an opportunity to examine how such institutional changes are associated with public discussions of LGBTQ+ topics. Here, using supervised and unsupervised natural language processing methods, we analysed 136,255 Facebook news posts and 65,969 public Instagram captions within a 100-day window before and after each reform. To contextualize online discourse with offline sentiment, we incorporated nationally representative survey data from the Gallup World Poll ($N = 46,200$). Following decriminalization, online discussions were characterized by reduced moralized and affectively negative language, particularly in contexts with greater freedom of expression. By contrast, offline perceptions of neighbourhood safety for same-sex couples shifted more modestly and unevenly across age groups and countries.

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LGBTQ laws and public opinion change

2026-06-10 · DOI 10.1038/S41562-026-02424-5

Yu-Te Huang

Abstract. Abstract unavailable.

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A reporting checklist for large language models in behavioural science

2026-06-09 · DOI 10.1038/S41562-026-02492-7

Stefan Feuerriegel; Christopher Barrie; M. J. Crockett; Laura K. Globig; Killian L. McLoughlin; Dan-Mircea Mirea; Arthur Spirling; Diyi Yang; Tim Althoff; Maria Antoniak; Lisa P. Argyle; Ashwini Ashokkumar; et al. (81 total)

Abstract. Abstract unavailable.

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Proceedings of the National Academy of Sciences

Conceptual priorities shape individual gaze patterns during naturalistic visual attention

2026-06-12 · DOI 10.1073/PNAS.2604369123

Amanda J. Haskins; Katherine O. Packard; Caroline E. Robertson

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Abstract. Our visual landscape consists of not only people, places, and objects (e.g., "soldier," "stadium," "flag") but also the conceptual relationships that unite them (e.g., "patriotism"). Because conceptual knowledge varies across individuals, this level of structure may support individualized patterns of attentional selection during naturalistic scene viewing. Here, we ask

whether individuals' gaze patterns reflect, in part, latent attentional priorities organized in conceptual space. Participants (N = 61) freely explored a diverse set of immersive real-world scenes (N = 100) in head-mounted VR while their gaze position was continuously recorded. We modeled gaze behavior using spatial, visual, and conceptual feature spaces, leveraging embeddings from large vision and language models, to uncover the latent priorities guiding individuals' unique patterns of selective attention across environments. Individuals exhibited stable and idiosyncratic gaze patterns across scenes and test-retest sessions, consistent with trait-like individual differences in attention. Spatial, visual, and conceptual feature spaces each explained unique variance in individual gaze patterns, with conceptual features contributing variance beyond that explained by spatial and visual features alone. Notably, language-model-based predictions were particularly effective at capturing these individualized patterns. Together, these findings indicate that naturalistic visual attention is structured at multiple levels-including a conceptual level-revealing stable individual differences in how people sample and prioritize information across complex visual environments.

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Misattributed biospheric carbon sink to the Three-North Program in the Taklamakan Desert may lead to an impractical policy: A commentary

2026-06-12 · DOI 10.1073/PNAS.2607916123

Tian Gao; Jiaojun Zhu; Xiao Zheng; Lining Song; Yirong Sun; Dexiong Teng; Xinhua Zhou

Campbell Scientific Inc.; Chinese Academy of Sciences Key Laboratory of Forest Ecology and Silviculture, Institute of Applied Ecology, Chinese Academy of Sciences; Chinese Academy of Sciences-Campbell Scientific Inc. Joint Laboratory of Research and Development for Monitoring Forest Fluxes of Trace Gases & Isotope Elements, Institute of Applied Ecology, Chinese Academy of Sciences; Qingyuan Forest Chinese Ecosystem Research Network, National Observation and Research Station

Abstract. Abstract unavailable.

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Adaptive self-organization of global swidden forests

2026-06-10 · DOI 10.1073/PNAS.2606552123

Sean S. Downey; Denis Tverskoi; Shane A. Scaggs; Xinyi Wu; Zaarah Syed; Jansen Lebowitz; Rongjun Qin; Stefan Thurner

Complexity Science Hub; Department of Anthropology, The Ohio State University; Department of Civil, Environmental and Geodetic Engineering, The Ohio State University; Department of Electrical and Computer Engineering, The Ohio State University; Department of Molecular Genetics, The Ohio State University; Division of Biostatistics, College of Public Health, The Ohio State University; Institute of the Science of Complex Systems, Center for Medical Data Science, Medical University of Vienna; Santa Fe Institute; et al. (11 total)

Abstract. Does swidden agriculture, a prototypical coupled human and natural system, exhibit a process of adaptive self-organization in which cultural practices balance environmental constraints through adaptive feedback? Here, we investigate whether quantitative signatures of adaptive self-organization can be detected in a dataset consisting of 18,000+ contiguous swidden patches in 18 remote sensing images of swidden mosaics from tropical and subtropical regions globally. We find that the distributions of patch sizes in 16 of 18 swidden areas exhibit power law patterns with scaling exponents 1, and correlation distances of 548 m. To account for these patterns, we develop a plausible ethnographically informed agent-based model of labor exchange, land use, and swidden site selection in which both sustainable and unsustainable resource uses can emerge out of interactions among individuals or households. By analyzing the model, we identify spatial synchronization of swidden sites as the driver of power law formation, while social norms of swidden labor can guide the system to an intermediate level of landscape disturbance. Both mechanisms are required to maintain harvests and ecosystem productivity at high levels. Our model advances theoretical understanding of the socioecological dynamics of swidden agriculture, and supports the hypothesis that adaptive self-organization may be a general characteristic of coupled human and natural systems.

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Bridging national climate targets and firm-level decisions: An agent-based model of China's steel decarbonization pathways

2026-06-08 · DOI 10.1073/PNAS.2528009123

Xiao Liu; Jinhong Du; Lingling Lv; Jingnan Hu; Xingang Liu

China National Environmental Monitoring Centre; School of Environment, Beijing Normal University; School of the Environment and Safety Engineering, Jiangsu University; State Key Laboratory of Environmental Criteria and Risk Assessment, Chinese Research Academy of Environmental Sciences

Abstract. China's steel sector is pivotal to national and global climate goals, yet decarbonization is constrained by entrenched coal-based processes and regional disparities. Top-down and bottom-up models often neglect firm-level heterogeneity and behavioral dynamics. Here, we develop an agent-based model linking policy signals-carbon caps, pollution standards, and resource limits-with firm-level investment decisions. We evaluate four scenarios combining current and strengthened climate and air targets with neutral or accelerated technology progress. All scenarios achieve deep CO₂ reductions (90 to 95% by 2060), but only the integrated policy-plus- technology pathway eliminates blast furnaces before midcentury, achieving the steepest declines. Policy- or technology-only pathways entail trade-offs, including lock-in to carbon capture and storage or delayed reductions. Our results show that synchronizing stringent climate policy with rapid technology cost declines is essential to steer China's steel industry toward a carbon-neutral trajectory.

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The Urban Pulse: Diagnosing the urbanization process as spiky, cyclical, and asynchronous

2026-06-08 · DOI 10.1073/PNAS.2537770123

Zhe Zhu; Michail Fragkias; Ji Won Suh; Ella McCoshan; Lelin Chen; Kexin Song; Juwon Kong; Tian Li; Karen C. Seto

Department of Economics, College of Business and Economics, Boise State University; Department of Geography, University of Victoria; Department of Natural Resources and the Environment, University of Connecticut; School of Integrative Plant Science, Cornell University; Yale School of the Environment, Yale University

Abstract. Many urban indicators predominantly focus on static outcomes, failing to capture the dynamic nature of urbanization. Here, we introduce the concept of the Urban Pulse to characterize urbanization as a process of continual spatiotemporal change. Using dense time series remote sensing data, we operationalize the Urban Pulse for six cities that represent diverse development levels, economies, and geographies: Seattle, Shenzhen, Lagos, Mumbai, Dubai, and Mexico City. We then use the Urban Pulse to assess how COVID-19 affected each of the six cities. The results empirically substantiate three major insights. First, urbanization is inherently spiky, characterized by episodic events that deviate from baseline trends. Second, urbanization is cyclical, reflecting transitions through stages of expansion, stabilization, or decline, and does not occur with predictable frequency, such as seasonal patterns. Third, urbanization is asynchronous, with subcity areas exhibiting uncoordinated development patterns. The Urban Pulse COVID-19 analysis shows that while urbanization processes for some cities have not returned to prepandemic patterns, other cities exhibit a significant shift in patterns postpandemic. These findings suggest that the selection of a neighborhood or subcity area as an observation unit may have little bearing on a city as a whole. By capturing these spatial and temporal rhythms, the Urban Pulse framework provides a comprehensive understanding of process dynamics, enabling researchers to link processes to outcomes more effectively. This approach allows for the recognition of early warning signals at subcity scales and may facilitate targeted interventions at the neighborhood level, where they are most impactful.

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Science

Patterns of brain-wide associations reflect socioeconomics

2026-06-11 · DOI 10.1126/SCIENCE.AEE6213

Scott Marek; Meghan Rose Donohue; Nicole R. Karcher; Caroline P. Hoyniak; Roselyne J. Chauvin; Ashley C. Meyer; John Miller; Andrew N. Van; Anxu Wang; Noah J. Baden; Vahdeta Suljic; Kristen M. Scheidter; et al. (36 total)

AI Institute for Health, Washington University School of Medicine, St. Louis, MO, USA.; Allied Labs for Imaging Guided Neurotherapies (ALIGN), Washington University School of Medicine, St. Louis, MO, USA.; Department of Biomedical Engineering, Washington University in St. Louis, St. Louis, MO, USA.; Department of Neurological Surgery, Washington University School of Medicine, St. Louis, MO, USA.; Department of Neurology, Washington University School of Medicine, St. Louis, MO, USA.; Department of Pediatrics, University of Minnesota Medical School, Minneapolis, MN, USA.; Department of Pediatrics, Washington University School of Medicine, St. Louis, MO, USA.; Department of Population and Public Health Sciences, Keck School of Medicine, University of Southern California, Los Angeles, CA, USA.; et al. (22 total)

Abstract. Previous brain-wide association studies (BWAS) have linked specific environmental and behavioral variables to brain variability. In this work, we mapped 649 variables to children's brains and compared the resultant BWAS maps with each other and with neurobiological reference patterns. Socioeconomic status (SES) showed the strongest brain-wide associations. The SES associations were strongest in motor and sensory but not cognitive regions, a pattern shared across many BWAS maps, including intelligence quotient (IQ). A single, common BWAS brain pattern existed across variables that was most reflective of a child's socioeconomics. Adjusting for SES weakened brain-IQ associations, eliminating the BWAS motor and sensory pattern. Brain-with-IQ associations also did not generalize when trained on higher-SES subsamples. Thus, children's brains vary the most with SES, potentially through SES-dependent sleep deprivation and stress.

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American Political Science Review

The Effects of Religious Messages and Endorsements on Political Attitudes: A Meta-Reanalysis

2026-06-10 · DOI 10.1017/S0003055426101695

RADHA SARKAR; ALEXANDER COPPOCK

Northwestern University; Tecnologico de Monterrey

Abstract. The experimental study of the effects of religious messages and endorsements by religious leaders on political attitudes has a relatively short history yet has coalesced around three main claims: religious treatments are especially effective because of their religious character, effects are larger for religious affiliates than nonaffiliates, and effects are larger for high religiosity types than low religiosity types. Here, we meta-reanalyze the experimental record (58 treatment-outcome pairs drawn from 43 studies reported in 26 papers) to probe the generalizability of these claims. Our findings indicate that these three headline claims do not generalize straightforwardly across contexts: effects are large in some cases and close to zero in others, and we find no evidence in favor of the claimed heterogeneities by religious affiliation or religiosity. Based on a census of the estimands in this literature, we offer suggestions for future research that would enhance commensurability and synthesis.

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The Domestic Sources of International Trust

2026-06-08 · DOI 10.1017/S0003055426101713

MICHAEL A. GOLDFIEN; MICHAEL F. JOSEPH; ROSEANNE W. MCMANUS

Pennsylvania State University; U.S. Naval War College; University of California, San Diego

Abstract. How do states overcome mistrust? Scholars argue that costly foreign policy signals build trust. But when trust is low, such as during rivalries, states are unwilling to use these signals for fear of being cheated. We argue that domestic policies can also build trust by revealing information about a state's likelihood of cooperating internationally when there is a correlation between domestic and international preferences. We further argue that domestic policies have a distinct advantage: the value states accrue from them depends less on international reciprocation. As a result, domestic choices can reassure counterparts at moments when trust is so low that costly international signals appear prohibitively risky. We test the

implications of our theory in case studies of the Cold War's end and United States-South Korea trust-building post-coup, illuminating several phenomena the current literature struggles to explain: initial trust-building between enduring rivals, asymmetric trust-building, and trust-building through illiberal domestic policies.

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The Expressive Duty to Vote

2026-06-08 · DOI 10.1017/S0003055426101725

CECILE FABRE

University of Oxford

Abstract. Standard arguments for the moral duty to vote in nationwide elections or referendums appeal to citizens' general duty of reciprocity not to free ride on the provision of the public goods of democratic governance and its benefits, to a nonreciprocal duty to maintain and strengthen democratic institutions, or to a general duty to bring about just outcomes. In this article, I argue that citizens are under an expressive duty to vote, as grounded in a more general duty to manifest support for just and democratic laws, and for democratic institutions as the instantiation of the requirement to treat one another with equal concern and respect. Even if (in circumstances I elucidate) citizens are morally permitted not to vote for either of the options on offer, it does not follow that they are morally permitted to abstain: sometimes, participating in the poll and voiding one's ballot is, on expressive grounds, the right thing to do.

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Nature

Deep learning four decades of human migration

2026-06-10 · DOI 10.1038/S41586-026-10611-7

Thomas Gaskin; Guy J. Abel

Abstract. Abstract Human migration is a fundamental driver of global demographic change, shaping population structure, labour markets and social policy across countries 1-3 . Although long-term migration patterns are often linked to economic development 4 , they can shift rapidly in response to shocks such as conflict, environmental crises and political change 5 . Despite its importance, migration remains difficult to measure consistently: existing data are sparse, concentrated in high-income settings and are fragmented across incompatible definitions, temporal resolutions and data types 6-8 . Past efforts have relied on partial datasets, including flow records, stock estimates and model-based reconstructions with limited coverage 9-14 . A central challenge is therefore to construct a globally consistent, high-resolution account of migration flows over time. Here we present a new dataset of annual origin-destination migration across 230 countries and regions from 1990 to the present, integrating diverse data sources into a unified modelling framework. By combining official statistics, census-based stocks, net migration estimates and past flow reconstructions, our approach produces temporally detailed and spatially comprehensive estimates that substantially extend existing resources. Using an ensemble of deep recurrent neural networks informed by geographic, economic, cultural and political covariates, we capture both persistent trends and short-term responses to changing conditions-all while propagating uncertainty to generate confidence bounds. Our results outperform existing five-year flow estimates on held-out data and provide finer temporal resolution, revealing previously obscured dynamics in global migration patterns. This framework highlights regions in which uncertainty remains high and data collection is most urgently needed. By releasing all data, code and trained models, we provide a transparent and reproducible foundation for future work. These advances enable a more timely and detailed understanding of human mobility, with implications for research and policy in an increasingly dynamic global system.

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Political Communication

Evaluating Echo Chambers, Rabbit Holes, and Radicalization Pathways on YouTube

2026-06-10 · DOI 10.1080/10584609.2026.2671765

Megan A. Brown; James Bisbee; Angela Lai; Richard Bonneau; Jonathan Nagler; Joshua A. Tucker

Firsthand AI, Firsthand AI, New York, New York, USA; Genentech / Roche; New York University, New York, New York, USA; University of Michigan, Ann Arbor, Michigan, USA; Vanderbilt University, Nashville, Tennessee, USA

Abstract. Abstract unavailable.

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Multi-Platform Referrers of Misinformation: A Comparative Analysis of Misinformation Visits Referred by Facebook, Twitter, Instagram, Reddit, YouTube, Snapchat, and TikTok

2026-06-09 · DOI 10.1080/10584609.2026.2679492

Ross Dahlke; Ryan C. Moore; Danya Adib-Azpeitia; Johan Ugander; Jeffrey T. Hancock

Stanford University; Stanford University, USA, Stanford, USA; The University of Texas at Austin, Austin, USA; University of Wisconsin-Madison; Yale University

Abstract. Journalistic and scholarly accounts alike often depict social media platforms as key players in exposing people to misinformation. Unfortunately, existing research examining social media's role in exposing people to misinformation online implicitly operates under what we call the direct referrer paradigm, focusing only on visits to misinformation that directly follow social media. We introduce the embedded referrer paradigm, which considers social media's more comprehensive role in leading people to visit misinformation throughout their online journeys and operationalize it via counterfactual simulations. We estimate the effects of social media in exposing people to misinformation websites under both paradigms using 3 months of a large sample of Americans' (N = 1,240) web browsing behavior (21 million website visits). We explore seven different platforms' (Facebook, Instagram, Reddit, Snapchat, TikTok, Twitter, YouTube) effects in exposing people to misinformation sites with various ideological leanings. Overall, our results suggest that analyses under the direct referrer paradigm underestimate the role of social media in exposing people to misinformation, with estimated effects under the embedded referrer paradigm ranging from 1.5 to 15+ times larger than effects under the direct referrer paradigm. The difference in estimated effects under both paradigms is generally consistent across platforms, misinformation types (liberal, conservative, non-ideological), and individuals' partisanship. However, for some combinations of platforms, types of misinformation, and mobile vs. desktop devices, the discrepancy between the two paradigms is even greater.

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Previous Week Catch-Up

Nature Communications

Brain neuromarkers predict self- and other-related mentalizing across adult, clinical, and developmental samples

2026-06-06 · DOI 10.1038/S41467-026-73945-W

Dorukhan Acl; Jessica R. Andrews-Hanna; Marina Lopez-Sola; Mariet van Buuren; Lydia Krabbendam; Liwen Zhang; Lisette van der Meer; Paola Fuentes-Claramonte; Edith Pomarol-Clotet; Raymond Salvador; Martin Debbane; Pascal Vrticka; et al. (19 total)

Abstract. Abstract Human social interactions rely on the ability to reflect on one's own and others' internal states and traits—a process known as mentalizing. Impaired or altered mentalizing is a hallmark of multiple psychiatric and neurodevelopmental conditions. Yet, replicable and easily testable brain markers of mentalizing have so far been lacking. Here, we apply an

interpretable machine learning approach to multiple datasets (total $n = 390$) to train and validate fMRI brain signatures that predict i) mentalizing about the self, ii) mentalizing about another person, and iii) both types of mentalizing. Self-mentalizing and other-mentalizing classifiers had positive weights in anterior/medial and posterior/lateral brain areas, respectively, with accuracy rates of 82% and 77% for out-of-sample prediction. The classifier trained across both types of mentalizing showed 98% predictive accuracy and separated (mental) attributional from factual inferences. Classifier patterns revealed better self/other separation in healthy adults compared to individuals with schizophrenia and with increasing age in adolescence. Together, our findings reveal consistent and separable neural patterns subserving trait-based mentalizing about self and others-present at least from the age of adolescence and functionally altered in severe neuropsychiatric disorders. These mentalizing signatures hold promise as candidate neuromarkers of social-cognitive processes in different contexts and clinical conditions.

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Science Advances

Bivalent impact of social networks on overarming: Insights on the alignment between social and individual interests

2026-06-05 · DOI 10.1126/SCIADV.AED3904

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Abstract. Drawing on evolutionary game theory, we present a stylized model of gun acquisition for individuals in a society where all have the right to bear arms. In our dynamic model, pairwise confrontations with attendant payoffs take place between individuals who are either armed or unarmed. Individual payoffs depend on the probability of confrontation, the choice to arm, and others' choices. We show that the likelihood of confrontation affects the optimal societal arming rate and the arming rate that occurs in equilibrium. The latter rate quickly surpasses the former one as the probability of confrontation increases, a phenomenon we call "overarming." This reflects a misalignment of individual and societal interests. We further show that spatial structures can exacerbate overarming, especially when individuals perceive a deteriorating social environment. Results from our base model and its extensions highlight the importance of understanding how fundamental behavioral dynamics and network heterogeneity influence individual decisions to acquire firearms, informing empirical research for mitigating overarming.

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Science

Fires reverse progress toward ozone air quality standards in the United States

2026-06-04 · DOI 10.1126/SCIENCE.AED3197

Weizhi Deng; Jun Wang; Meng Zhou; Xi Chen; Xiaodong Wu; Huanxin Zhang; Jason B. Cohen; Jing Wei; Arlindo da Silva; Guy P. Brasseur; Claire Granier; Laurence Rouil

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Abstract. Recent surges in wildfire emissions have exacerbated surface ozone pollution in the United States. Using deep learning, we developed a gapless daily surface ozone dataset at 1-kilometer resolution for 2003-2024. This dataset revealed a reversal in national policy-relevant ozone trends that had gone undetected by the sparse monitoring network: from 0.65 parts per billion (ppb) per year (2003-2015) to +0.13 ppb per year (2015-2024). The reversal was primarily driven by increasing wildfire emissions, offsetting 3.9 years of mitigation progress. Premature deaths from fire-sourced ozone have increased by 318 deaths per year since 2013, with post-2013 mortality 46% higher than pre-2013 mortality. During 2022-2024, wildfire emissions exposed 43 million people to nonattainment conditions, effectively preventing a 4-ppb tightening of the ozone standard. These results underscore the growing challenges of sustaining air quality progress as wildfires intensify under climate change.

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Home alone: Remote work, isolation, and mental health

2026-06-04 · DOI 10.1126/SCIENCE.AEC7671

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Abstract. How does remote work affect isolation and mental health? We drew on five nationally representative surveys of American workers (N = 588,322) conducted from 2011 to 2024, omitting the peak pandemic years of 2020-2021. Our difference-in-differences approach compared changes in mental health among people in remotable jobs—who experienced a large and persistent rise in remote work since COVID-19—to people in nonremotable jobs, where remote work increased far less. We found that remote work increases time spent alone, worsens mental well-being across multiple measures, and increases the use of mental health services and prescriptions. These effects were concentrated among individuals living alone. We estimate that the rise of remote work explains about a third of the increase in isolation and mental distress between 2011-2019 and 2022-2024.

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Landscape efficiency frontiers for biodiversity, climate mitigation, and net economic value

2026-06-04 · DOI 10.1126/SCIENCE.AEA9058

Stephen Polasky; Peter L. Hawthorne; Rebecca Chaplin-Kramer; Jeffrey Smith; James S. Gerber; Saleh Mamun; Mary Ruckelshaus; Jason Russ; Rafael Schmitt; Adrian L. Vogl; Adam C. Castonguay; James Douglass; et al. (30 total)

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Abstract. National governments and multilateral institutions face difficult challenges reconciling biodiversity, climate, and economic development goals. We integrated spatial biophysical and economic data with optimization methods to develop sustainable landscape efficiency frontiers that show maximally feasible combinations of biodiversity conservation, land-based climate mitigation, and net economic value from agricultural crops, livestock, and forestry production. We applied this approach in 146 countries and found large potential gains in biodiversity, climate, and economic development from improved land use and land management. Summing national-level results shows the potential to increase climate mitigation by more than 200 billion metric tons of CO₂ equivalents (>20% increase) or net economic value by more than US\$350 billion (>80% increase), without loss in other objectives.

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The architecture of the internet creates risks for democracy

2026-06-04 · DOI 10.1126/SCIENCE.AEI2409

Stephan Lewandowsky

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Abstract. Abstract unavailable.

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American Journal of Political Science

The demand for elections under autocracy: Regime approval and the elimination of local elections in Russia

2026-06-03 · DOI 10.1111/AJPS.70060

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Abstract. Abstract Most contemporary autocracies hold elections. Does the public value these elections? If so, do they value them enough to punish incumbents that subvert elections? We examine this question in contemporary Russia, examining whether individuals withdraw support from regime leaders when local elections are eliminated. Over the past 25 years, most Russian cities have replaced their directly elected mayors with appointed executives. This paper uses the largest data set on public opinion ever assembled on Russia-over 400,000 responses drawn from two decades by Russia's top polling agencies-to analyze how the elimination of elections in Russia's large cities has affected public attitudes toward the authorities. Using a difference-in-differences design, we find that election cancellation reduces support for President Vladimir Putin. This effect is stronger in settings with histories of robust electoral competition. This suggests that the public is more likely to punish incumbents for eliminating elections when individuals have experience with competitive elections.

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Nature

Can an army of babies and dogs rescue psychology from its reproducibility crisis?

2026-06-03 · DOI 10.1038/D41586-026-01727-X

Nic Fleming

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The future of science communication is not an article like this

2026-06-02 · DOI 10.1038/D41586-026-01723-1

Abstract. Abstract unavailable.

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Political Analysis

Post-Treatment Problems: What Can We Say about the Effect of a Treatment among Sub-Groups Who (Would) Respond in Some Way?

2026-06-02 · DOI 10.1017/PAN.2026.10047

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Abstract. Abstract Investigators are often interested in how a treatment affects an outcome for units responding to treatment in a certain way. We may wish to know the effect among units that, for example, meaningfully implemented an intervention, passed an attention check or demonstrated some important mechanistic response. Simply conditioning on the observed value of the post-treatment variable introduces problematic biases. Further, the identification assumptions required by several existing strategies are often indefensible. We propose the treatment reactive average causal effect (TRACE), which we define as the total effect of treatment in the group that, if treated, would realize a particular value of the relevant post-treatment variable. By reasoning about the effect among the "non-reactive" group, we can identify and estimate the range of plausible values for the TRACE. We demonstrate the use of this approach with three examples: (i) learning the effect of police-perceived race on police violence during traffic stops, a case where point identification may be possible; (ii) estimating effects of a community policing intervention in Liberia, in communities that meaningfully implemented it; and (iii) studying how in-person canvassing affects support for transgender rights, among participants for whom the intervention would result in more positive feelings toward transgender people.

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Proceedings of the National Academy of Sciences

Bilingualism predicts executive function resilience after COVID-19 in aging

2026-06-01 · DOI 10.1073/PNAS.2532470123

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Abstract. Bilingualism has been associated with enhanced executive functions (EFs), particularly attentional control, and may confer protection against cognitive decline in older age. At the same time, the COVID-19 pandemic has emerged as a factor negatively affecting EF in older adults. Bilingualism might offer resilience against these COVID-related cognitive declines, especially in late adulthood, by bolstering cognitive reserve. The present study collected data from 312 community-dwelling individuals spanning the adult lifespan (aged 18 to 80) to address two goals. Goal 1 was to identify a latent structure linking bilingualism and EF across the adult lifespan, using a continuous multivariate approach to help resolve controversies in the literature. Goal 2 was to determine whether bilingualism can protect against post-COVID EF decline in the most vulnerable older age group. We used partial least squares correlation (PLSC) analysis to extract latent relationships between multiple bilingual experience measures (and relevant covariates) and a battery of EF task outcomes. Results revealed that individuals with greater bilingual experience (earlier L2 age of acquisition and greater L2 proficiency) showed a multivariate association with executive function measures, with the most consistent behavioral expression observed in task-switching performance. Earlier L2 AoA was most robustly associated with better nonverbal task-switching performance among older adults reporting a history of COVID, even when modeling COVID burden continuously (i.e., number of infections and recency of last infection). These findings demonstrate that bilingual experience contributes to resilience in attentional control under flexible task-switching demands in older adults even in the context of COVID-19.

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Climate change, inequality, and childhood stunting in African countries

2026-06-01 · DOI 10.1073/PNAS.2518179123

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Abstract. Childhood stunting is associated with increased mortality, higher risk of chronic disease, impaired cognitive development, lower educational attainment, reduced economic opportunities, and intergenerational transmission of stunting. These risks are likely to intensify as climate change exacerbates key drivers of undernutrition, making it important to understand how rising temperatures affect stunting and the role of socioeconomic inequality in this relationship. We analyze data from 34 African countries from 2004 to 2020 using Demographic and Health Surveys (DHS), combining observed temperature variability from ERA5 reanalysis with anthropogenic temperature anomalies derived from Detection and Attribution Model Intercomparison Project (DAMIP) simulations. Using a mediation framework, we examine pathways linking temperature anomalies, inequality, and child stunting. Results based on observed temperature anomalies show no direct climate effect on stunting or inequality, but a positive and statistically significant association between inequality and stunting. In contrast, anthropogenic temperature anomalies are positively associated with inequality and stunting, with a 1 C increase linked to a 3.45% rise in child stunting (SE = 1.52, P = 0.023), with no evidence of mediation through inequality. Notably, we find a consistent positive association between inequality and stunting across specifications. These findings suggest that reducing inequality, together with investments in education, sanitation, and household resilience, could substantially lower stunting rates and protect child health in a warming world.

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Egress thresholds and wildfire fatalities

2026-06-01 · DOI 10.1073/PNAS.2535081123

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Abstract. Avoiding human fatalities during wildfires is a key public policy objective. Outward road access, or the number of egress routes, is widely assumed to influence wildfire fatalities, yet few studies have quantified if or when this factor becomes critical. To address this gap, we assembled a dataset on community-level wildfire fatality counts and combined it with nationally consistent community egress for the United States, finding that cumulative fatalities are sharply concentrated in communities with very few exits, declining steeply to roughly six nonresidential roads, beyond which additional routes confer minimal further risk reduction. Extending this analysis nationally, we mapped all small communities (

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Impacts of future sea level change on Greenland from community knowledge, coastal mapping, and glacial isostatic adjustment models

2026-06-01 · DOI 10.1073/PNAS.2528615123

Kirsty J. Tinto; Jacqueline Austermann; Robin E. Bell; David Blockley; Casey E. Brayton; Diana Krawczyk; Lauren Lewright; Andrew J. Lloyd; Frank O. Nitsche; Guy J. G. Paxman; David F. Porter; Aqqaq Srensen; et al. (14 total)

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Abstract. Making predictions of global change relevant to individual communities is a critical challenge, especially where local impacts diverge from global predictions. While sea level is rising globally, in Greenland, sea level is predicted to fall over the coming century due to land rebound and sea surface lowering, consequences of glacial isostatic adjustment from both past and future ice loss. Here we present results of a US and Greenland scientific partnership, working with the community of Aasiaat, West Greenland, to identify the impacts of future sea level change where sea level is projected to fall 0.41 to 0.93 m

by 2100 CE. Through collaboratively designed, conducted, and interpreted multibeam bathymetry surveys combined with targeted glacial isostatic adjustment models, we identify areas that will be most impacted by sea level fall. We identify two shallow channels where disruption to navigation and nutrient cycles will occur when water level falls beneath critical depth thresholds. To make these predictions useful we established communication pathways to share findings effectively across a broad array of interest groups by engaging with community members, government officials, and students. This multipronged approach of collaboratively designed observations, locally targeted impact modeling and integrated formal and informal communication can be expanded to coastal communities facing environmental challenges around the world.

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Resetting population mobility responses under repeated nonpharmaceutical interventions: Implications for hypothesized pandemic fatigue

2026-06-01 · DOI 10.1073/PNAS.2533284123

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Abstract. Nonpharmaceutical interventions (NPIs) are essential for controlling infectious diseases during prevaccine periods, yet their success hinges on sustained population behavioral responses to restrictions. This study investigates how mobility responses to tiered restriction systems evolved over time during COVID-19 in six geographical regions across Europe, North America, Africa, and South America. Using daily mobility data and linear-mixed models, we characterized three temporal dimensions shaping these responses: cumulative exposure to restrictions, time spent under a given tier, and repeated reintroduction of the same tier. Time spent under a given tier caused the most rapid attenuation of mobility reduction, producing effects within days that cumulative exposure, required months of restrictions to achieve. Iterative application of shorter NPIs, interspersed with temporary relaxation, was associated with a restoration of mobility responses when restrictions were reintroduced, mitigating attenuation and sometimes even strengthening responses. These patterns are consistent with mechanisms discussed under the umbrella of pandemic fatigue, whereby temporary relaxation may provide psychological relief and a sense of regained autonomy that can renew public willingness to respond to restrictions when reintroduced, while remaining compatible with behavioral adjustment and structural constraints. However, they are also compatible with alternative explanations, including behavioral adaptation, structural constraints on mobility, or evolving interpretations of restrictions. These findings emphasize the dual benefits of short, strategic NPIs for epidemic control and public resilience, offering actionable insights for designing more sustainable pandemic interventions.

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Resolving Feynman's restaurant problem reveals optimal solutions and human strategies

2026-06-01 · DOI 10.1073/PNAS.2509612123

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Abstract. In the 1970s, physicist Richard Feynman turned lunch with a friend into a math problem-how to optimize dish selection over multiple meals-but his handwritten notes remained a mystery for decades. Here we present the fully deciphered problem and solution, prove its optimality, generalize it to related problems, and compare the results to human behavior. The optimal policy specifies decreasing thresholds for switching from exploring new dishes to exploiting the best, with thresholds varying based on the distribution of the quality of dishes. We connect these results to the existing psychological literature on optimal stopping problems, which has explored close variants on Feynman's problem, and use our generalization of the solution to explore how the underlying distribution of the quality of the options influences people's

choices. A preregistered experiment with 2,520 participants shows that people adopt thresholds that decrease linearly with the proportion of trials remaining, consistent with the observation of linear thresholds in other optimal stopping problems. However, we show that people tend to explore more than predicted by linear thresholds, and that different distributions of quality result in thresholds with the same slope but different intercepts. These results indicate that people adapt linear thresholds used in optimal stopping tasks in a way that is sensitive to the underlying distribution—a simple strategy that we show is nearly as effective as Feynman's solution.

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Rethinking energy transition strategies for the European Union amid rising energy prices

2026-06-01 · DOI 10.1073/PNAS.2609606123

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Abstract. Geopolitical tensions and the disruption of major energy imports have sharply elevated energy prices across the European Union (EU), creating a persistent energy supply gap and posing long-term challenges for energy security. Here, we introduce a comprehensive framework integrating the advanced Greenhouse Gas and Air Pollution Interactions and Synergies (GAINS) model with a system dynamic WILIAM ("Within limits") Integrated Assessment Model to systematically evaluate strategies for closing this energy gap. We find that measures to close the energy gap in the short term exacerbate either economic or public health burdens, highlighting the need for more sustainable solutions in the long term. Our further analysis shows that long-term strategies focused on renewable energy transitions not only close the energy gap but also bring net benefits equivalent to ~3% of projected GDP for 2050. Moreover, we find that rising energy prices substantially influence the cost-effectiveness of accelerated energy transitions. Under current high energy price levels, the benefits from achieving energy targets set for 2050 by 5 to 20 y in advance could outweigh the additional costs. The consideration of dynamic trends in the levelized cost of electricity (LCOE), particularly under rapid renewables expansion, is critical for avoiding overly optimistic estimations of green transition benefits. Our findings suggest that a timely transition to renewable energy benefits the EU in the face of the energy crisis, and policies proactively adjusting energy prices could facilitate a greener, more sustainable future.

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The 2025-2030 Dietary Guidelines for Americans are associated with higher land, water and nitrogen use, and greenhouse gas emissions

2026-06-01 · DOI 10.1073/PNAS.2604814123

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Abstract. Dietary guidelines form the foundation of federal food policy, informing programs such as the Supplemental Nutrition Assistance Program (SNAP), government food procurement, school meals, and military and hospital food services. The most recent guidelines, released in early 2026, advise avoiding "hyperprocessed" (generally known as ultraprocessed foods, UPFs), while also recommending higher overall protein intake chiefly from animal sources. Here, we quantified the greenhouse gas (GHG) emissions, freshwater use, land use, and nitrogen fertilizer use of adherence to these new guidelines in comparison to the current Mean American Diet (MAD). We constructed isocaloric diets with varying degrees of added protein intake and protein sources that are similar to current dietary patterns, but with no UPFs as recommended. We find

that although eliminating ultraprocessed foods from the MAD could yield substantial environmental benefits-given their large share of total intake and animal-based makeup-the recommended increase in animal-based proteins, will more than offset this environmental dividend in GHG, land use, and nitrogen fertilizer use by as much as 32%, but not in water use (i.e., water use is lower by 7 to 19% than the MAD). Across all environmental metrics measured, favoring plants- over animal-based foods leads to better environmental outcomes. Overall, adherence to the 2025-2030 Dietary Guidelines for Americans would confer net environmental harm across most key metrics relative to the prevailing U.S. diet.

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